

Headphone Output Stage (OPA1622): Working Document

Digital Audio Player • Rev 4

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1. Summary

Field	Value
Amplifier	Texas Instruments OPA1622IDRCR, dual, 10-pin VSON 3 × 3 mm, thermal pad to V-
3.5 mm jack	Same Sky SJ1-3535NS: stereo TRS, right angle, through hole, shielded, tip and ring switches
ESD protection	Texas Instruments ESD122DMXR, 2-channel bidirectional, on tip and ring
Function	Unity-gain headphone driver, one non-inverting follower per channel, with output-inclusive compensation
Signal chain	I/V → differential summing and MFB low-pass → OPA1622 follower → 3.5 mm jack

The summing stage already sets the level (0.85 V peak) and polarity, so the output stage only has to deliver current into the IEM with very low output impedance and stay stable into cable capacitance. A follower keeps polarity, barely loads the MFB filter, and applies the full loop gain as feedback. A small isolation resistor R_{iso} separates the op-amp from cable capacitance; audio-frequency feedback (R_{fb}) is taken after R_{iso} so the loop removes its effect, and high-frequency feedback (C_f) is taken before it so the loop does not see the capacitive load.

2. Design Requirements

Requirement	Value
Gain	1
Load	Simgot EW300, 28 Ω $\pm$ 15% (23.8 Ω worst case): $\approx$ 36 mA peak per channel at 0.85 V peak

Requirement	Value
Output impedance	< 1 Ω , 20 Hz to 20 kHz
Coupling	DC coupled
Stability	Stable into cable capacitance
DC-fault protection	Firmware monitors the LT3042 and LT3093 power-good outputs and drives EN low on a rail fault. The LT3042 PG goes high impedance in shutdown, so a complete loss of the +3V9 feed is not flagged by that PG.
Filtering	None added. The I/V pole and MFB filter already give $\approx$ 60 dB at 1 MHz.
Supply	$\pm$ 3.3 V analog rails
Constraints	Low noise · Small footprint · COG capacitors in the signal path

3. Pin Configuration

3.1 OPA1622IDRCR

Pin	Name	Connection
1	+IN A	SUMM_R
10	-IN A	R_{fb} to the ring node; C_f to OUT A
9	OUT A	R_{iso} to the ring node (RIGHT_EAR)
5	+IN B	SUMM_L
6	-IN B	R_{fb} to the tip node; C_f to OUT B
7	OUT B	R_{iso} to the tip node (LEFT_EAR)
2	V+	+3V3_ANALOG, 0.1 μ F + 1 μ F to GND
4	V-	-3V3, 0.1 μ F + 1 μ F to GND
3	GND	GND (reference for EN)
8	EN	Enable divider from the EN_HP GPIO (Section 5)
Pad	Thermal pad	V- (-3.3 V), own pour and via field

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#), Section 5 (Pin Functions).

3.2 SJ1-3535NS (3.5 mm jack)

Pin	Function	Connection
1	Sleeve	GND
2	Tip	Left output (TIP): ESD122 IO2, 10 k Ω bleed to GND
3	Ring	Right output (RING): ESD122 IO1
4	Tip switch (closed to tip, opens on insertion)	Detect node: 100 k Ω pull-up to +3V3_DIG, 10 k Ω series to PLUG_DET (GPIO)
5	Ring switch	Not connected
SH1 to SH3	Shield	GND

Source: [SJ1-353XNS Series Datasheet \(Same Sky\)](#), mechanical drawing and pin table.

4. External Components

Ref	Value	Part / Notes	Qty
U	Dual op-amp	TI OPA1622IDRCR, 10-pin VSON	1
J	3.5 mm jack	Same Sky SJ1-3535NS	1
D	2-channel ESD	TI ESD122DMXR, bidirectional, 0.27 pF, ±3.6 V I/O range	1
R_{iso}	4.7 Ω	Op-amp output → jack node	2
R_{fb}	499 Ω , 0.1%, 0402	YAGEO RT0402BRD07499RL (#1-105). Jack node → -IN.	2
C_f	1.5 nF, C0G, 1%, 0603	Murata GCM1885C1H152FA16D (#1-106). Op-amp output → -IN.	2
C_{dec}	0.1 μ F X7R + 1 μ F X5R	One pair per supply rail	4
R_{EN1}	54.9 k Ω	EN_HP GPIO → EN	1
R_{EN2}	50 k Ω	EN → GND; holds EN low at boot	1
C_{EN}	0.1 μ F X7R	EN → GND; enable ramp	1
R_{DET}	100 k Ω	Jack detect node → +3V3_DIG	1
R_{SER}	10 k Ω	Detect node → PLUG_DET; limits current if the tip swings negative	1
R_{BLEED}	10 k Ω	TIP → GND; defines the tip voltage when the amp is disabled	1

Plug detect: with no plug, the tip switch ties the detect node to the tip, held near 0 V by the amplifier or R_{BLEED} , so PLUG_DET reads low (≤ 0.3 V). Inserting a plug opens the switch and PLUG_DET reads 3.3 V. The detect node must not share the PLUG_DET net name, or R_{SER} is bypassed.

Tip bleed: the OPA1622 output is not guaranteed high impedance in shutdown, so R_{BLEED} defines the tip voltage for detection. It adds 10 k Ω in parallel with a 23.8 Ω IEM (≈ 85 μ A at 0.85 V peak, 0.24% of the load current), is linear, and sits inside the feedback loop, so it does not change output impedance or distortion.

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#), Section 7.4.1 (Shutdown Mode); [ESD122 Product Page and Datasheet \(Texas Instruments\)](#).

5. Equations and Results

Feedback handover frequency

$$f_x = \frac{1}{2\pi R_{fb} C_f} = \frac{1}{2\pi \cdot 499 \Omega \cdot 1.5 \text{ nF}} = 213 \text{ kHz}$$

Output impedance

$$Z_{out}(f) \approx R_{iso} \left| \frac{R_{fb}}{R_{fb} + Z_{C_f}} \right|, \quad Z_{C_f} = \frac{1}{j 2\pi f C_f}$$

$$Z_{out}(20 \text{ kHz}) \approx 4.7 \Omega \cdot \frac{499}{|499 - j 5305|} = 0.44 \Omega$$

Source: voltage division between R_{fb} and C_f in the two-path feedback network; valid while loop gain is high (well below the 8 MHz unity-gain bandwidth).

Enable divider (EN_HP GPIO at 3.3 V)

$$V_{EN} = 3.3 \text{ V} \cdot \frac{50 \text{ k}\Omega}{54.9 \text{ k}\Omega + 50 \text{ k}\Omega} = 1.57 \text{ V} > V_{IH,max} = 0.95 \text{ V}$$

$$\tau_{EN} = (54.9 \text{ k}\Omega \parallel 50 \text{ k}\Omega) \cdot 0.1 \mu\text{F} = 2.6 \text{ ms}$$

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#), Section 6.5 (Enable pin: V_{IL} 0.82 V typical, 0.95 V maximum over -40 to 125 °C; V_{IL} 0.78 V typical) and Section 7.3.3 (thresholds referenced to the GND pin; EN limited to (V-) - 0.5 V to (V+) + 0.5 V).

Plug-detect level, unplugged and amplifier off

$$V_{DET} = 3.3 \text{ V} \cdot \frac{R_{BLEED}}{R_{DET} + R_{BLEED}} = 3.3 \text{ V} \cdot \frac{10 \text{ k}\Omega}{110 \text{ k}\Omega} = 0.30 \text{ V}$$

Quantity ($R_{iso} = 4.7 \text{ }\Omega$)	Value
Z_{out} at 20 kHz	0.44 Ω
Extra swing across R_{iso} at 36 mA peak	0.17 V (op-amp peak $\approx$ 1.02 V)
R_{fb} thermal noise, 20 Hz to 20 kHz	2.9 nV/ $\sqrt{\text{Hz}}$, $\approx$ 0.4 μV RMS

5.1 Stability

Datasheet phase margin for a follower ($G = +1$): $\approx 70^\circ$ at 200 pF, $\approx 45^\circ$ at ≈ 430 pF, $\approx 30^\circ$ at 600 pF, $\approx 0^\circ$ at 1 nF. Simulation used a 10 mV step with C_{load} swept at 300 pF, 600 pF and 1 nF.

Load	Network	Overshoot at 300 pF / 600 pF / 1 nF
23.8 $\Omega \parallel C_{load}$	2 Ω / 499 Ω / 1.5 nF	Clean at all three
10 k $\Omega \parallel C_{load}$	Plain follower	Much worse than the compensated network
10 k $\Omega \parallel C_{load}$	2 Ω / 499 Ω / 1.5 nF	$\approx 10\%$ / 20% / 40%
10 k $\Omega \parallel C_{load}$	4.7 Ω / 499 Ω / 1.5 nF	$\approx 10\%$ / 15 to 20% / 30%, settles in 1 to 2 cycles

The 23.8 Ω IEM load damps cable capacitance. 10 k $\Omega \parallel 1$ nF is a harsh bound (line-out into a long cable), outside the intended use.

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#), Figure 24 (Phase Margin vs Capacitive Load).

6. Important Values

Item	Value
Output swing from rail	800 mV (no load), 900 mV (600 Ω): $\approx \pm 2.4$ V on ± 3.3 V
Common-mode input range	(V-) + 1.5 V to (V+) - 1 V: -1.8 V to +2.3 V
Gain-bandwidth	8 MHz at $G = 1$ (32 MHz at $G = 1000$)
Quiescent current	2.6 mA per channel; 5 μA per channel in shutdown
Input voltage noise	2.8 nV/ $\sqrt{\text{Hz}}$ at 1 kHz
EN thresholds	High ≥ 0.82 V typ (0.95 V max); low ≤ 0.78 V typ (0.65 V min over temperature)
Jack rating	1 A, 12 V DC; 5,000 cycles

Source: [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#), Section 6.5; [SJ1-353XNS Series Datasheet \(Same Sky\)](#).

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-25	Initial issue: topology, pin connections, component values, EN network, stability data.
1	2026-09-25	R_{iso} placed; plug detect finalized.
2	2026-09-25	Thermal pad, detect net label, output impedance and EN limit resolved.
3	2026-09-25	Reformatted; part numbers clarified; plug-detect section folded into components; EN threshold and tip-bleed sources added; equations typeset.
4	2026-09-26	DC-fault protection set to firmware monitoring of LDO power-good with EN shutdown; open problem closed.

9. Sources

1. [OPA1622 Datasheet, SBOS727B \(Texas Instruments\)](#)
2. [SJ1-353XNS Series Datasheet \(Same Sky\)](#)
3. [ESD122 Product Page and Datasheet \(Texas Instruments\)](#)